RAWDet-7: A Multi-Scenario Benchmark for Object Detection and Description on Quantized RAW Images

Anonymous authors
Paper under double-blind review

Abstract

Most vision models operate on 8-bit standard RGB (sRGB) images produced by dedicated image sensor processing pipelines designed for human perception rather than machine reasoning. In contrast, RAW images preserve sensor measurements, dynamic range, and fine-grained scene structure that can be critical for downstream understanding. Yet, progress in the RAW-domain vision remains limited by the lack of large-scale, high-quality benchmarks. Thus, we introduce RAWDet-7, a multi-scenario benchmark for object detection and object description on quantized RAW images, comprising ∼25k training and 7.6k test images consolidated from four datasets across diverse cameras, sensors, bit-depth precision, lighting conditions, and environments. RAWDet-7 provides dense, standardized annotations for seven object categories, correcting missing labels, hallucinations, and inconsistencies in prior datasets, especially for small and partially-occluded instances. Beyond detection, we introduce an object-description track with detailed object-level descriptions derived from high-resolution sRGB references, enabling the study of how well different RAW processing pipelines preserve fine-grained semantic, spatial, and contextual information. Finally, RAWDet-7 supports controlled benchmarking under simulated 4-bit, 6-bit, and 8-bit quantization. Across standard detectors and a large grounding model, we show that suitable RAW-aware input mappings make low-bit RAW on par with, and in some settings superior to, sRGB pipelines, while preserving substantially richer object-level descriptions than naïve quantization. Dataset & code upon acceptance.

1 Introduction

Modern vision systems are typically built and evaluated on standard RGB (sRGB) images at 8-bit precision. At capture time, however, image sensors record the scene in a substantially higher bit-depth (∼24-bit) RAW format, which is then mapped to sRGB by an image signal processor (ISP). This mapping consists of multiple stages, including black-level subtraction, demosaicking, denoising, white balancing, gamma correction, color correction, and compression, and is primarily designed to produce visually pleasing images for humans. However, this photography-oriented pipeline is not optimized for downstream machine perception.

Compared to processed sRGB images, RAW images retain more faithful sensor measurements, including the original bit depth and dynamic range. Leveraging RAW data can therefore benefit downstream vision tasks (Omid-Zohoor et al., 2014; Xu et al., 2023). At the same time, RAW data varies substantially across sensors, which differ in bit depth, response characteristics, and ISP design. For specialized applications (Klinghoffer et al., 2022; Sommerhoff et al., 2023; Agarwal et al., 2025), jointly optimizing sensing, image processing, and the downstream task can therefore be highly beneficial. Despite this promise, progress in RAW-domain vision and end-to-end learned task-specific imaging remains limited by the lack of large-scale, well-annotated, and standardized RAW benchmarks.

To address this limitation, we introduce RAWDet-7, a multi-scenario benchmark for object detection and object description on RAW images. We focus on object detection because it requires both recognition and precise localization of class-specific visual evidence. In addition, we introduce an object-description track on a curated subset of images to study how much fine-grained semantic, spatial, and contextual

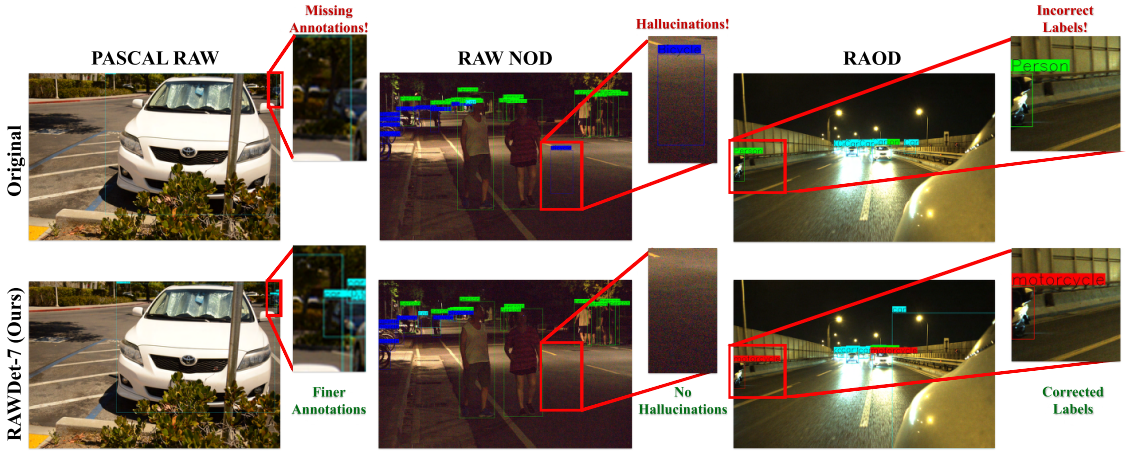

Figure 1: Comparing ground truth annotations provided in the original datasets and the new ones proposed in RAWDet-7. Our proposed annotations are more fine-grained, as seen for PASCAL RAW, which originally annotated only one instance of the cars in the image; we annotate all the other instances of cars in the image (with a 20% overlap threshold i.e. at least 20% area of the bounding box should be non-overlapping). RAW NOD (-Nikon and -Sony), RAOD (-Day and -Night) original annotations contain hallucinations as seen here for RAW NOD, which hallucinates a bicycle in the center right of the frame. Original annotations even contain misclassifications, as seen for RAOD, which misclassified a motorcycle as a person. RAWDet-7 **(bottom) overcomes these drawbacks.**

information remains accessible after RAW preprocessing and low-bit quantization. The proposed RAWDet-7 is larger than previous RAW detection datasets, comprising ∼25k training and ∼7.6k test images obtained by consolidating four different datasets (Omid-Zohoor et al., 2014; Xu et al., 2023; Morawski et al., 2022; Ignatov et al., 2020) across varied lighting conditions, camera types, and environments.

Many prior RAW datasets cover only a small number of classes and often annotate only large, prominent objects. We therefore relabel the consolidated dataset using a large foundation model, followed by human validation on the corresponding sRGB images, to obtain annotations for 7 object categories aligned with the community-accepted naming conventions of MS-COCO (Lin et al., 2014) and LVIS (Gupta et al., 2019). This produces substantially denser annotations, both by increasing category coverage and by improving the labeling of small and partially-occluded instances, thereby making RAWDet-7 more useful for developing robust detection models that generalize across real-world scenarios. As illustrated in Fig. 1, RAWDet-7 provides significantly richer and corrected annotations than prior datasets for the same images. Since annotation quality is central to the value of the benchmark, we further validate the reliability of the relabeling pipeline through human studies, confirming that the resulting annotations are consistently preferred over the previously existing ones.

While RAWDet-7 contains full-precision RAW images, reducing bit depth can lower memory usage and provide significant power savings (Yin et al., 2021). However, there is still limited understanding of how complex downstream vision models behave under heavily quantized RAW input. We therefore benchmark object detection performance under simulated low-bit quantization, in addition to standard sRGB input. Specifically, we consider low-bit quantized inputs obtained using different information processing methods: linear scaling, logarithmic scaling (Buckler et al., 2017), learnable γ scaling (Ljungbergh et al., 2023), and a combination of logarithmic and learnable γ scaling (Fatima et al., 2025).

Through systematic benchmarking on low-bit quantized RAW inputs, we show that models trained on RAWDet-7 outperform counterparts trained on the individual source subsets across quantization levels, with more robust detections for small and occluded instances, further demonstrating the value of the benchmark. Moreover, essentially all prominent object detection methods considered here, including their architectural and preprocessing design choices, were developed for sRGB input. Despite this clear bias, our experiments show that object detectors trained on RAW input, especially low-bit RAW input with suitable

input scaling, can perform on par with, and in some settings surpass, models trained on the same amount of sRGB data.

Beyond detection, we evaluate how well different RAW processing pipelines preserve object-level information for a large vision-language model trained primarily on sRGB images. To this end, we use detailed descriptions generated from high-resolution sRGB images as reference descriptions, and compare them against descriptions obtained from different processed RAW variants of the same content. This setup does not treat sRGB captions as absolute semantic ground truth; rather, it measures how closely different RAW representations preserve the information that an sRGB-biased model can extract from its native input regime. We further validate this object-description pipeline through human studies, showing that the generated descriptions are of high quality and that LLM-based judgments are meaningfully aligned with human ratings. Our results show that descriptions generated from suitably processed RAW images are substantially closer to the sRGB references, and provide more accurate and detailed object explanations, than descriptions obtained from naïvely processed low-bit RAW images.

The contributions of this work are as follows:

- **A consolidated RAW benchmark.** We introduce RAWDet-7, a large-scale benchmark for object detection and object description on quantized RAW images, built by consolidating four datasets and re-annotating them into seven standardized categories with denser labels, especially for *small* and *partially-occluded* objects. Thereby, the object-description track allows assessing how well RAW preprocessing preserves fine-grained semantic, spatial, and contextual information.
- **Human-validated benchmark quality.** We validate the benchmark with human studies both for detection and description annotations. Users prefer our consolidated detection annotations over the original ones and confirm the quality of the object-description annotations.
- **Controlled low-bit RAW benchmarking.** We benchmark downstream vision under simulated 4-bit, 6-bit, and 8-bit RAW quantization with four input scaling strategies, and show that suitable RAW-aware mappings make low-bit RAW competitive with, and sometimes better than, standard sRGB pipelines. We show that these findings often hold across models and tasks, in particular across both conventional object detectors and a recent large grounding model.

2 Related Work

In the following, we discuss prior works towards object detection on RAW images, and relevant RAW image datasets that align closely with the proposed RAWDet-7.

Object detection. Object detection has progressed from region-based methods to one-stage detectors and, more recently, to large-scale open-set grounding models. Among contemporary approaches, foundation-model-based detectors such as **MM-Grounding-DINO** (Zhao et al., 2024) extend detection beyond fixed label spaces and enable open-set localization from language supervision. This recent line of work is particularly relevant for our study, since it allows us to examine whether observations on RAW and quantized RAW inputs also transfer to modern large-scale detection models. Earlier deep object detectors are commonly grouped into two-stage and one-stage paradigms. Two-stage methods, such as **Faster R-CNN** (Ren et al., 2016), first generate region proposals and then classify and refine them, forming a strong and widely used detection baseline. One-stage methods perform detection in a single pass, often improving efficiency while maintaining competitive accuracy. In this category, **RetinaNet** (Lin et al., 2017) addresses class imbalance through focal loss, while **PAA** (Kim and Lee, 2020) improves anchor-based detection through probabilistic anchor assignment. Together, these detectors span representative and influential design paradigms, making them well suited for evaluating detection performance in the RAW image domain.

Object Description. Recent works have explored set-of-mark and visual prompting strategies to improve region-level grounding and object-centric captioning in large vision-language models (Yang et al., 2023a; Wan et al., 2024; Lei et al., 2025; Cai et al., 2024; Yang et al., 2023b). Set-of-Mark prompting overlays alphanumeric marks and segmentation-derived regions so that a VLM can reason about specific objects and

their relations directly on the image (Yang et al., 2023a). (Wan et al., 2024) instead compares predictions on images with highlighted versus blacked-out regions to reduce prior biases and better follow region prompts. (Lei et al., 2025) introduces a coordinate grid on the image together with textual references to enhance spatial reasoning and vision-language coordination. ViP-LLaVA (Cai et al., 2024) trains multimodal models to interpret arbitrary user-drawn prompts such as boxes, arrows, and scribbles as region selectors, while (Yang et al., 2023b) uses precise segmentation masks and blur-outside-mask operations to focus the model on a target instance. In RAWDet-7, we adopt a set-of-marks interface and additionally provide a high-level object class for each numeric mark when prompting the VLM, allowing us to study how well detailed object descriptions are preserved across different quantizations.

RAW Imaging Datasets. Most existing datasets and models for object detection are developed on RGB images produced by ISP pipelines optimized for human perception. In contrast, RAW images preserve richer scene information and higher dynamic range, which may benefit high-level tasks such as detection (Ljungbergh et al., 2023; Xu et al., 2023; Wu et al., 2024). However, large-scale annotated RAW datasets remain scarce. Existing detection datasets like RAOD (Xu et al., 2023), NOD (Morawski et al., 2022), MultiRAW (Li et al., 2024b) and PASCAL RAW (Omid-Zohoor et al., 2014) are limited in either the number of images, object classes, or annotation quality, PASCAL RAW annotates only large objects, NOD and RAOD contain hallucinated or noisy labels, and all three cover a narrow range of conditions. The Zurich dataset (Ignatov et al., 2020), originally designed for RAW-to-RGB ISP learning, lacks detection labels and is re-annotated in our work. Other RAW datasets (Ignatov et al., 2020; Schwartz et al., 2019; Dang-Nguyen et al., 2015) focus on ISP or forensics, not object detection.

To address these limitations, prior works propose synthetic RAW data generation (Li et al., 2024a; Zhang et al., 2024) or task-adaptive ISP modules (Cui and Harada, 2024; Diamond et al., 2021; Mosleh et al., 2020; Robidoux et al., 2021) for improving detection. Some methods (Buckler et al., 2017; Xu et al., 2023) evaluate the effect of low-bit quantization on recognition but do not optimize quantization-aware pipelines, while others (Ljungbergh et al., 2023; Liu et al., 2024a) jointly learn simplified ISPs with detection models. In contrast, our proposed dataset, RAWDet-7, unifies and re-annotates existing datasets with fine-grained labels for seven categories across varied lighting, sensor types, and scenes, and enables controlled benchmarking under realistic 4-bit, 6-bit, and 8-bit quantization. For evaluation, we adopt standard detectors including Faster R-CNN (Ren et al., 2016), RetinaNet (Lin et al., 2017), and PAA (Kim and Lee, 2020), as well as MM-Grounding-DINO (Zhao et al., 2024), demonstrating that our findings generalize across architectures and detection paradigms.

3 Proposed Dataset RAWDet-7

RAWDet-7 provides high quality annotations for object detection as well as for object description. The following sections provide the details of the annotation process.

3.1 Object Detection Annotations

As discussed in Sec. 1, progress in RAW-domain vision has been limited by the lack of large-scale, high-quality benchmarks and by the practical challenges of handling RAW data. Although several RAW object detection datasets (Omid-Zohoor et al., 2014; Morawski et al., 2022; Xu et al., 2023) have been proposed, we discussed in Sec. 2 that they each suffer from practical limitations. For example, PASCAL RAW (Omid-Zohoor et al., 2014) and NOD (Morawski et al., 2022) focus on only a few coarse object classes, restricting annotations largely to 'Car', 'Person', and 'Bicycle'. As shown in Fig. 1, several annotations are missing even within these classes, particularly for small or partially occluded people and vehicles. Existing datasets also contain annotation errors and hallucinations, e.g., in Fig. 1, a motorcycle (under the original 'vehicle' category) from RAOD (Xu et al., 2023) is mislabeled as a 'Person'. Beyond annotation quality, sensor-specific differences in bit depth, sensor characteristics, and lighting conditions introduce substantial variability that is only partially captured by previous datasets. MultiRAW (Li et al., 2024b) attempts to broaden class and scenario coverage, but introduces contradictory categories such as 'rider' and 'person' while still remaining too small overall for a stable and standardized benchmark.

Table 1: Comprehensive comparison of datasets in RAWDET-7 and other datasets.

Dataset	Sensor Model	Resolution	Bit Depth	Scenarios	Train Images	Test Images	# Classes	# Sensors
PASCAL RAW (Omid-Zohoor et al., 2014)	Nikon D3200 DSLR	6034 × 4012	12	Day	2128	2130	3	1
Zurich (Ignatov et al., 2020)	MP Sony Exmor IMX380	2944 × 3958	10	Day	-	-	-	1
Raw-NOD-Nikon (Morawski et al., 2022)	Nikon D750	3968 × 2640	14	Night	3373	843	3	1
Raw-NOD-Sony (Morawski et al., 2022)	Sony RX100 VII	4256 × 2848	14	Night	3309	828	3	1
RAOD (Xu et al., 2023)	Sony IMX490	2880 × 1856	24	Day & Night	16089	4000	5	1
MultiRAW (Li et al., 2024b)	Multiple	Various	10, 12, 14, 24	Day & Night	5154	2315	10	5
RAWDET-7 **(Ours)**	Multiple	Various	10, 12, 14, 24	Day & Night	**24864**	**7688**	7	5

To address these limitations, we introduce RAWDET-7, a comprehensive benchmark dataset for object detection on RAW images. Comprising more than 32k RAW images, with ∼25k for training and 7.6k for testing, RAWDET-7 consolidates and standardizes four existing datasets: PASCAL RAW (Omid-Zohoor et al., 2014) (12-bit), RAOD (Xu et al., 2023) (24-bit, including day and night HDR scenes), Zurich RAW (Ignatov et al., 2020) (10-bit), and NOD (Morawski et al., 2022) (14-bit, with Sony and Nikon sensors). Together, these sources cover a wide range of imaging conditions, including daytime, nighttime, and HDR scenes, as well as diverse camera hardware and sensor bit depths. By unifying them under a consistent annotation scheme, RAWDET-7 enables controlled studies of generalization across lighting conditions, scene complexity, and sensor variability.

A central contribution of RAWDET-7 is its improved and standardized labeling. To correct missing annotations, hallucinations, and inconsistencies in the source datasets while establishing a uniform labeling protocol, we re-annotated all datasets using Grounded-DINO 1.5 (Liu et al., 2024b) (paid API version at: `https://deepdataspace.com/request_api`) for seven object categories: *'Car'*, *'Truck'*, *'Tram'*, *'Person'*, *'Bicycle'*, *'Motorcycle'*, and *'Bus'*, following MS-COCO and LVIS-style naming conventions. We then refined the generated annotations by retaining only confident predictions above a confidence threshold of 0.8. This threshold was chosen after manual inspection of annotation quality and accuracy over a large subset of RAWDET-7. As a result, RAWDET-7 provides denser and more consistent annotations across the consolidated dataset, with substantially improved coverage of small and occluded object instances. The improvement is not only qualitative: in an annotation-quality user study with 53 participants and 1325 pairwise comparisons, users preferred the RAWDET-7 annotations over the original annotations in 1002 cases, i.e., 75.62% of all comparisons. Tab. 1 lists the properties of the individual subsets of RAWDET-7 and MultiRaw (Li et al., 2024b).

3.2 Object Description

While object detection provides precise localization, it offers only a coarse, class-level characterization of scene content. To evaluate how much fine-grained information remains accessible after RAW preprocessing, especially for the annotated object categories of interest, we introduce the task of *object description*. The goal is not to define absolute semantic ground truth, but to measure how well different processed RAW variants preserve the object-level semantic, spatial, and contextual information that a strong vision-language model can extract from its native sRGB input regime.

To this end, we generate high-quality reference descriptions using Gemini-2.5-Pro (Comanici et al., 2025) for objects localized by the ground-truth bounding boxes in RAWDET-7. Specifically, for a curated subset of 500 high-resolution sRGB images, we overlay visual marks on annotated objects following the set-of-marks procedure of (Yang et al., 2023a), using black square markers with colored numbers indicating object IDs. We then prompt Gemini-2.5-Pro to produce detailed descriptions for each marked object. To account for output variability and to establish a reference agreement level, we prompt the model twice per image. Our prompts, qualitative examples, and human-study validation are provided in Sec. H, where a caption-quality user study with 63 participants and 1260 ratings assigns the generated object descriptions an average score of 4.07 ± 0.48 out of 5, i.e., approximately 82%, indicating that the reference descriptions are of high quality even under human evaluation.

For evaluation, we compare descriptions generated from the high-resolution sRGB images against those generated from downsampled sRGB variants and differently processed RAW variants at multiple bit depths, using the same set-of-marks. We report RegeX overlap and BLeU score, and additionally use Gemini-

2.5-Flash-Lite as a judge to score semantic similarity, level of detail, and precise detail match between the reference sRGB descriptions and those from each variant. Together, these metrics quantify how well different processing pipelines preserve object-level information beyond class labels alone.

4 Scaling Methods Used For Benchmarking

One key challenge with using RAW images in machine learning is that their high precision and large size, make them computationally expensive to store and process. Quantization offers a practical solution by reducing bit depth, but it introduces new design choices around how to scale inputs before quantization. As an example use-case, in this section, we explore combinations of quantization with known input scaling methods, such as logarithmic and gamma mappings, to enable efficient and effective object detection directly on low-bit RAW inputs. Recording at lower bit-depth and skipping the ISP before feeding images to a downstream task model can provide power and memory gains (Yin et al., 2021), making such methods practical in scenarios like low-bandwidth or low-power consumption sensing.

We benchmark the performance of RAWDET-7 using low-bit depth images obtained using the following scaling methods, where $\mathcal{X}$ with bit-depth N is the input to the quantizer $\mathcal{Q}(\cdot)$, which provides output with target bit depth $\hat{N}$:

1. **Linear quantization** converts input into a quantized digital value using uniform linear steps as

$$\mathcal{Q}(\mathcal{X}) = \left\lfloor \mathcal{X}_{\text{norm}} \cdot (2^{\hat{\text{N}}} - 1) \right\rfloor, \tag{1}$$

   where, $\mathcal{X}_{\text{norm}} = \left(\frac{\mathcal{X}}{2^{\text{N}}-1}\right)$ is the normalized version of $\mathcal{X}$. This naïve approach of reducing the bit-depth often fails, motivating better methods for scaling.

2. **Logarithmic quantization** proposed by (Buckler et al., 2017; Bermak and Kitchen, 2006) non-linearly scales the dynamic range of $\mathcal{X}$ before quantization,

$$\mathcal{Q}(\mathcal{X}) = \left\lfloor \frac{\mathcal{X}_{\log} - \min(\mathcal{X}_{\log})}{\max(\mathcal{X}_{\log}) - \min(\mathcal{X}_{\log})} \cdot (2^{\hat{\text{N}}} - 1) \right\rfloor, \tag{2}$$

   where, $\mathcal{X}_{\log} = \log(\mathcal{X} + \epsilon)$, with ϵ is accounting for the mapping to issue values $\mathcal{X}_{\log} \geq 0$, i.e., (Buckler et al., 2017) use $\epsilon = 1$.

3. **Quantization with Learnable γ -Scaling:** γ mapping, similar to logarithmic quantization, also compresses the dynamic range of the input, and we consider a learnable γ mapping before quantization as

$$\mathcal{Q}(\mathcal{X}, \gamma) = \left\lfloor \mathcal{X}_{\text{norm}}^{\gamma} \cdot \left(2^{\hat{\text{N}}} - 1\right) \right\rfloor, \tag{3}$$

   where $\mathcal{X}_{\text{norm}}$ is the normalized version of $\mathcal{X}$ as defined earlier in linear quantization, and the parameter γ is learned along with the parameters θ of the neural network. In addition to learning a single γ for the dataset, we also evaluate conditioning γ on the domain, such as lighting conditions or sensors. $\mathcal{Q}(\mathcal{X}, \gamma)$ is hereby referred to as γ-SCALING.

4. Since various non-linear mappings of input intensities are an active area of research for the sensors community (Gulve et al., 2023; Lefebvre and Bol, 2025), we benchmark a combination of logarithmic and learned γ quantization. That is, we use $\mathcal{X}_{\text{norm}} = \mathcal{X}_{\log}$ in Eq. (3).

4.1 Learning Task-specific Low-bit Quantization

We learn the parameter γ of γ-SCALING and 'Log + γ-SCALING' jointly with the neural network parameters for the specific task by optimizing

$$\min_{\theta,\gamma} \; \mathbb{E}_{(\mathcal{X},\text{y})} \left[\mathcal{L}(\eta(\mathcal{Q}(\mathcal{X}, \gamma)), \text{y}; \theta) \right], \tag{4}$$

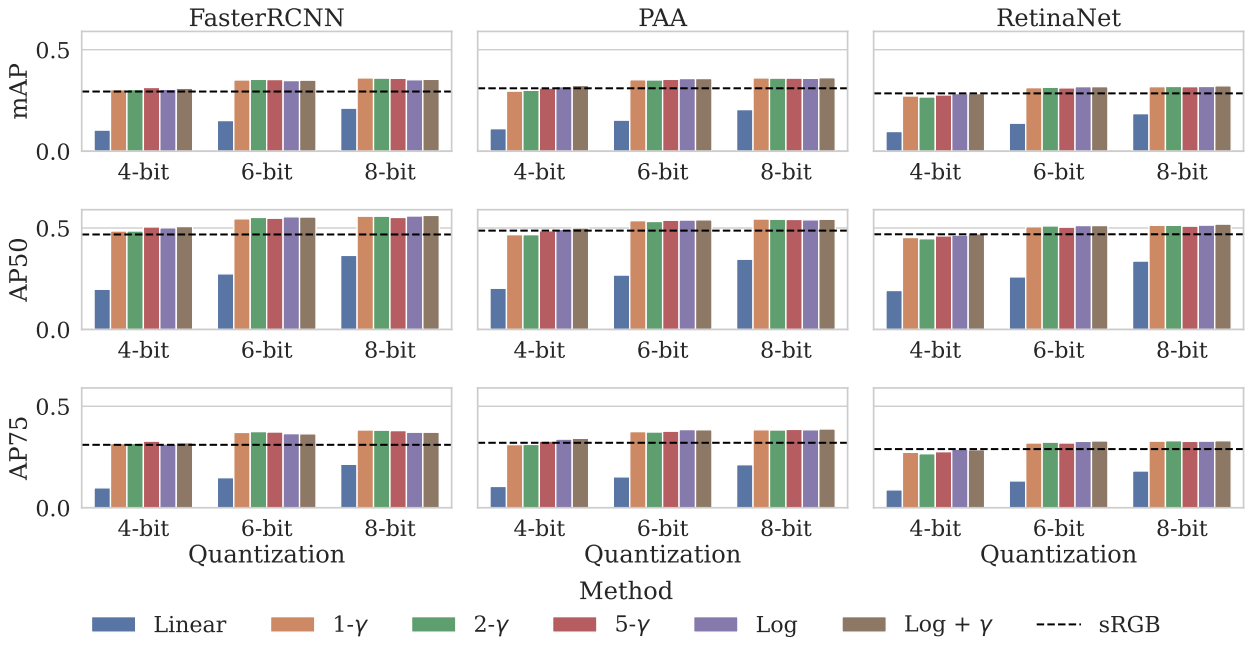

Figure 2: Benchmarking performance on RAWDet-7. Baselines such as logarithmic quantization and jointly learnt γ improve results across quantization levels and architectures.

where η is the task-specific neural network with parameters θ, and where $\mathcal{L}$ is a suitable loss function comparing the network output and the ground truth prediction y.

Since RAWDet-7 offers a unique multi-scenario setting, we also learn scenario-specific γ values. Using one γ value over the entire dataset is denoted as 1-γ. When using two γ values we denote it as 2-γ, with the intention of them conditioning to the time of day, i.e. one γ for daytime and one for nighttime images. Lastly, we test using five γ values, denoted as 5-γ, allowing one optimized to each sensor.

For optimization, since the quantization operation stops the flow of gradients in the backward pass, we use a straight-through estimator to allow gradient-based optimization of γ. Please note that $\gamma < 0$ would lead to undesirable behavior, i.e. , $\mathcal{X}_{\mathrm{Q}} \geq 2^{\hat{\mathrm{N}}} - 1$ when $\mathrm{N} \geq \hat{\mathrm{N}}$ (which is the case when we want to quantize to lower bit depths). To avoid this, we clamp γ to be non-negative, using ReLU(.) (Glorot et al., 2011). To take into account Bayer-patterned images in the RAW dataset, we extract red, green, and blue channels from the Bayer-patterned image and downsample it using nearest neighbor interpolation to simulate the capture of a low-resolution, low-bit-depth image, which is provided as input to the neural network. This pipeline for joint training of γ and task-specific neural network parameters θ using quantized RAW images is the methodology for the forthcoming evaluations.

5 Experiments

For object detection we use the mmdetection (Chen et al., 2019) framework from openmmlab. Following previous works (Xu et al., 2023; Omid-Zohoor et al., 2014) that perform object detection on RAW images, we conduct experiments using three architectures Faster R-CNN (Ren et al., 2016), RetinaNet (Lin et al., 2017), and PAA (Kim and Lee, 2020) models with a pre-trained ResNet50 (He et al., 2016) backbone. For fair comparison with sRGB counterpart, we extract the 'R'(Red), 'G' (Green), and 'B'(Blue) channels from the Bayer patterned RAW input, average over the two 'G' channels and report the following quantitative evaluation metrics: mean average precision (mAP) is calculated by averaging the Average Precision (AP) values across the intersection over union (IoU) thresholds ranging from [0.5, 0.95] with a step size of 0.05,

Table 2: Experiments with large VLM, MM-Grounding-DINO (Zhao et al., 2024) with a Swin-T (Liu et al., 2021) backbone. For experiments with γ-SCALING and Log + γ-SCALING we finetune only the γ-SCALING parameter while keeping the VLM frozen, for the other methods, we evaluate zero-shot.

Quantization	Method	mAP	AP50	AP75	AP95
RGB	sRGB	0.188	0.215	0.204	0.088
4-bit RAW	Linear	0.047	0.060	0.053	0.008
	Log	0.114	0.138	0.127	0.034
	γ-SCALING	0.104	0.125	0.115	0.035
	Log + γ-SCALING	0.135	0.161	0.149	0.048
6-bit RAW	Linear	0.066	0.081	0.074	0.014
	Log	0.149	0.182	0.164	0.042
	γ-SCALING	0.177	0.206	0.193	0.072
	Log + γ-SCALING	0.198	0.230	0.217	0.085
8-bit RAW	Linear	0.074	0.091	0.083	0.017
	Log	0.151	0.186	0.167	0.041
	γ-SCALING	0.177	0.207	0.193	0.070
	Log + γ-SCALING	0.202	0.233	0.220	0.090
12-bit RAW	Linear	0.075	0.092	0.084	0.017
	Log	0.152	0.187	0.168	0.041
	γ-SCALING	0.167	0.196	0.183	0.066
	Log + γ-SCALING	0.201	0.232	0.219	0.089

resulting in 10 threshold values. We also report performance with IoU-thresholds of 0.5 and 0.75 (AP50 and AP75) for each case. Results with 4 channels (R,G,G,B) are given in Sec. G, see Tab. 4. We use an ImageNet-1k (Russakovsky et al., 2015) pretrained backbone and freeze the first stage of the backbone during training. We follow the multi-scale training setup commonly used (Ren et al., 2016; Lin et al., 2017), during training, and at test time, the shorter edge length is kept at 800. We use a batch size of 16 and train the model for 140 epochs. We use SGD with Nesterov Momentum (Sutskever et al., 2013) as the optimizer with a weight decay of $1e^{-3}$. For the PASCAL RAW dataset, we use the same train/test split as proposed by the original paper. For the Nikon, Sony, and Zurich datasets, we do an 80-20 train/test split of the entire dataset. For RAOD, we use the publicly available validation set as the test set. All dataset splits and statistics are provided in Fig. 13.

5.1 Benchmarking Performance on RAWDET-7

We report the performance of various object detection networks in the proposed RAWDET-7 benchmark, for different quantization levels in Fig. 2. Here we observe that naïvely reducing the bit depth through linear scaling and quantization to a low bit depth degrades the quality of fine-grained details that are critical for tasks like object detection. In comparison, the non-linear scaling and quantization methods perform significantly better at all the bit depths. Optimizing only a single additional parameter γ (for γ -scaling) jointly with the downstream network during training, we observe clear improvements over linearly scaled and quantized inputs, and competitive performance relative to sRGB in several settings. This shows that even a lightweight, task-aware mapping of the input for quantization can lead to meaningful improvements over naïve quantization pipelines. Logarithmic scaling and quantization, albeit fixed, perform on par with or better than learned γ scaling. Among the different target bit depths considered, we observe that 4-bit linear scaling and quantization perform worse across all architectures. Improvements are observed by applying a logarithm or the learned γ scaling before the quantization. The improvements are on par with the sRGB setting, indicating that simple pre-quantization mappings such as log or learned γ can substantially reduce the gap to sRGB-based pipelines in this benchmark, and work towards alleviating the need for an ISP. We also learn separate γ during training for day and night images as well as separate γ for different sensors indicated by 2-γ and 5-γ, respectively in Fig. 2. We observe a marginal difference in performance while training with multiple γ values, while combining logarithmic scaling with learned γ scaling provides some gain in detection performance when compared to solely using logarithmic scaling for networks trained from scratch.

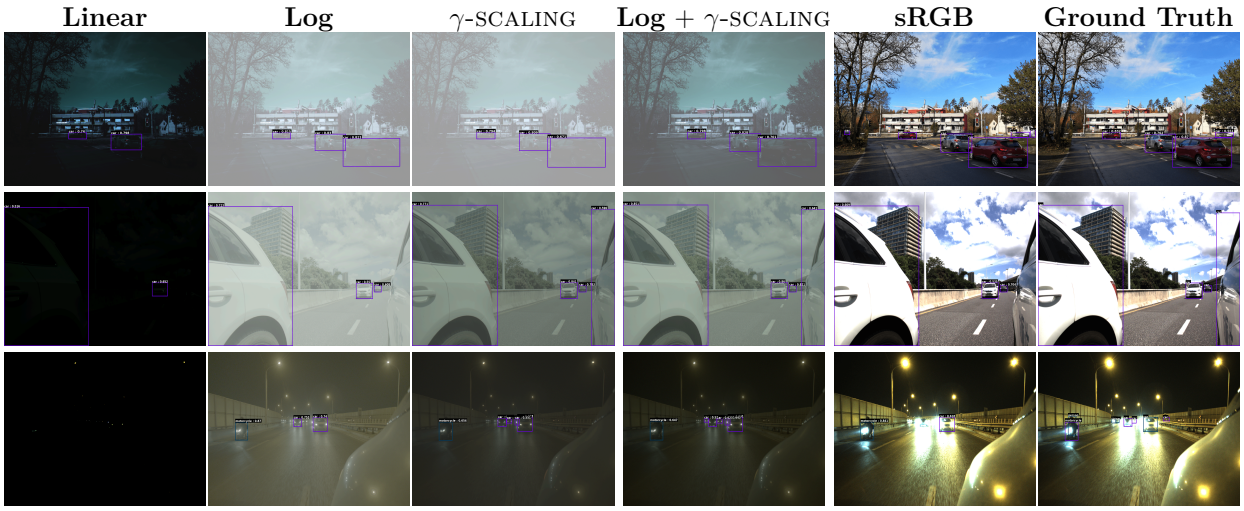

Figure 3: Visualizing predictions made by MM-Grounding-DINO on RAWDET-7 at 6-bit quantization, using different methods for predictions on sRGB images and the expected Ground Truth predictions. Her,e for methods γ and Log + γ, we finetune only the γ parameter while keeping the VLM frozen; for the other methods, we evaluate zero-shot. We randomly sample three distinct scenarios to demonstrate the versatility of the proposed RAWDET-7.

5.2 Benchmarking Large VLMs (LVLMs)

We also benchmark RAWDET-7 on MM-Grounding-DINO for 4 quantization levels and various signal scaling techniques. Unfortunately, training or fine-tuning the VLM is computationally infeasible given our limited resources. Thus, we keep the VLM frozen throughout. Linear, log, and sRGB are evaluated zero-shot, while for γ-scaling and log+γ-scaling we optimize only the scalar γ parameter. Thus, for these, we keep the VLM frozen and only train the γ value with lr 1e-5 and a maximum training budget of 50 epochs. We report these results in Tab. 2 and observe that Log + γ-SCALING performs the best, even exceeding the performance on sRGB for some quantization levels. Log + γ-SCALING highlights the details of the images in a better way than other image-enhancing baselines. These results are promising, as they show that under this evaluation protocol, suitably processed RAW inputs can yield higher detection performance than sRGB on RAWDET-7 for some quantization levels (6-bit quantization and above), even with a model trained on a very large amount of sRGB data. Since at 6-bit depth itself, RAW input to MM-Grounding-DINO outperforms sRGB input, we visualize qualitative results for 6-bit depth quantization in Fig. 3.

5.3 Object Description

Next, we study object-description quality across different RAW preprocessing pipelines. Since current closed VLMs cannot be fine-tuned end-to-end on RAW data, we adopt a zero-shot setting and reuse the γ values that showed strong robustness in the object detection experiments across architectures and bit depths. We randomly sample 500 images from the RAWDET-7 validation set and treat high-resolution sRGB as the reference modality. Each image is annotated with set-of-marks and provided twice to Gemini-2.5-Pro (Comanici et al., 2025) to assess the stability of object descriptions on sRGB.

Unlike object detection annotations, which provide precise localization but only coarse class labels, object descriptions allow us to evaluate how much object-level semantic, spatial, and contextual information is preserved under RAW preprocessing. They therefore provide a consistent reference for controlled, fine-grained comparisons across preprocessing methods and image variants.

From the corresponding Bayer-patterned RAW measurements, we extract R, G, and B channels and construct quantized RAW variants at 4, 6, and 8 bits using two mappings: linear and log + γ, where the latter combines logarithmic compression with the reused γ values. We additionally generate a 2× downsampled sRGB image

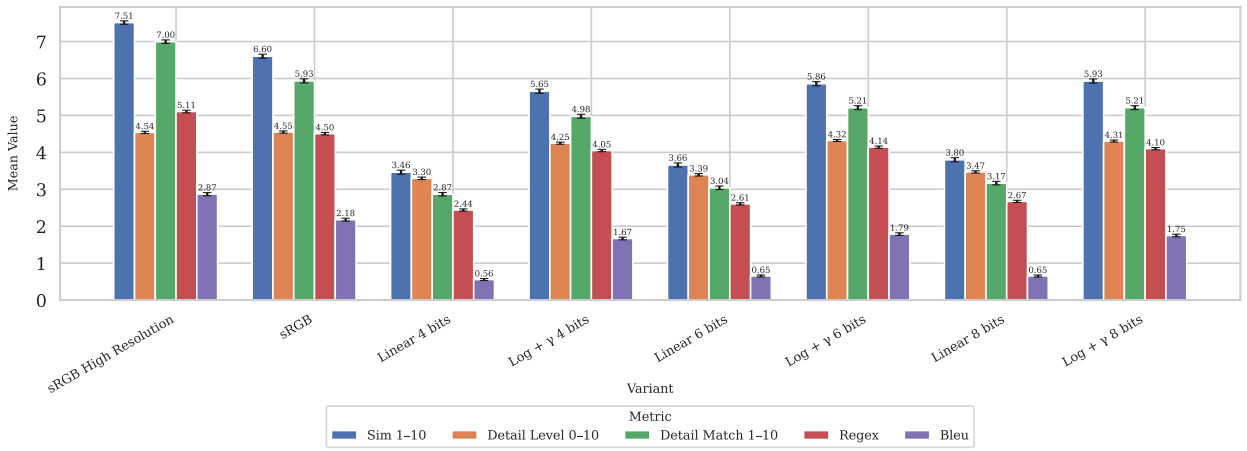

Figure 4: Object Description quality across different image variants. Descriptions are generated from linear and log+γ variants of quantized images using Gemini, with reference captions obtained from higher-resolution sRGB images. Here, sRGB refers to the sRGB images downsampled 2× to match the resolution of the RAW images after extracting R, G, and B channels. We multiply BleU and Regex scores by 10 to align them with the other metrics. Black lines indicate standard errors. The results show that captions generated from processed RAW images (log+γ) achieve competitive quality, closely matching the reference captions.

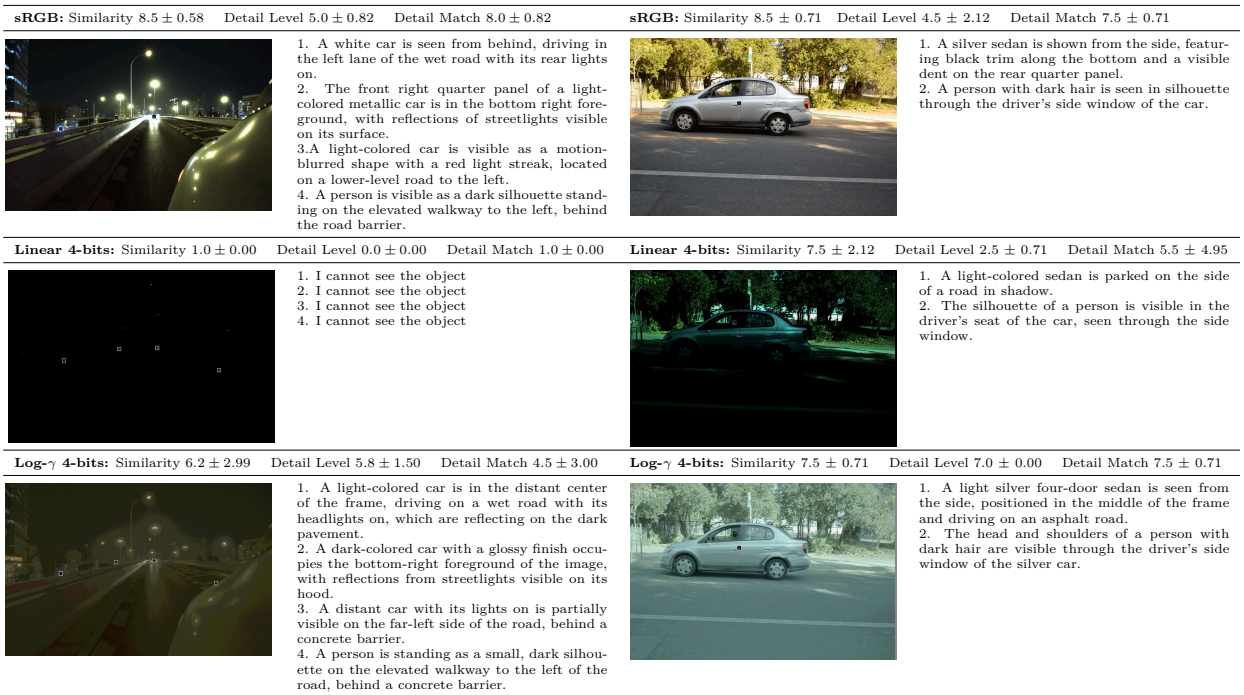

Figure 5: Comparison of caption quality for different image representations, including quantitative scores. Captions generated from Log-γ images provide more detail-level explanations compared to those from Linear 4-bit quantized images.

whose effective resolution roughly matches that of the 8-bit RAW variants. All non-reference representations, namely downsampled sRGB, linear RAW, and log + γ RAW, are overlaid with the same set-of-marks and passed to Gemini-2.5-Pro for object descriptions. Caption quality is evaluated against the high-resolution sRGB reference using five metrics: Bleu, Regex Match, Similarity (1-10), Detail Match (1-10), and Detail

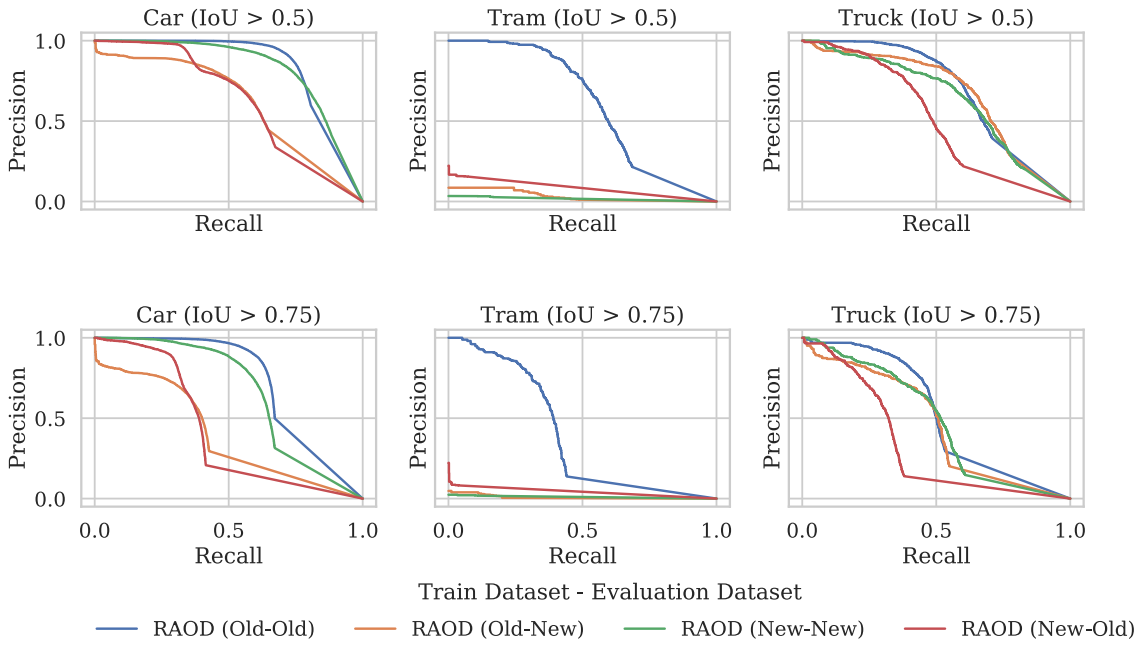

Figure 6: Precision Recall Curves for RAOD. "Old" refers to the annotations as proposed in the original dataset, whereas "New" means the annotations in RAWDet-7. All the models are trained on FasterRCNN, for 8-bit quantization and jointly learnt with 1 gamma. As mentioned in the legend, the keys are "(Train Dataset - Evaluation Dataset)".

Level (1-10), where the last three are scored by Gemini-2.5-Flash-Lite for consistent qualitative assessment. Formal metric definitions are deferred to the appendix Sec. H.3.

The results in Fig. 4 show a clear and consistent pattern. Across all bit depths, linearly quantized RAW images produce captions that are less similar to the sRGB reference and substantially worse in detail and detail match, indicating that naive quantization removes semantically important structure. In contrast, log + γ quantized RAW images preserve much more fine-grained information and achieve scores close to, and sometimes approaching, high-resolution sRGB, while the 2× downsampled sRGB variant lies between these extremes. This supports our central claim that careful RAW preprocessing can substantially narrow the gap between what an RGB-pretrained VLM can describe on sRGB and on RAW inputs. At the same time, the remaining gap and the strong sensitivity to preprocessing highlight that poor RAW processing leads to systematically worse descriptions, whereas good RAW-aware mappings are feasible and worth optimizing, as illustrated in Fig. 5. The object-description track in RAWDet-7 makes these trade-offs measurable and enables future work to benchmark RAW preprocessing pipelines not only against sRGB, but also across multiple bit-depth-accurate RAW variants. Looking ahead, optimizing γ specifically for object description may further improve performance.

5.4 Discussion: Added Value by the Contributed Annotations and Descriptions

Fig. 6 shows the precision-recall curves for the RAOD dataset in four different settings. "Old-Old" means that the model was trained with original annotations and tested with them as well, whereas "Old-New" indicates a different test setting, testing with newly proposed annotations, and likewise for "New-New" and "New-Old". Since the annotations in both cases are slightly different, we only compute the precision-recall curves for 4 mutually common annotations i.e. *'Car'*, *'Tram'*, *'Truck'*, and *'Person'*. We observe that the precision when tested with new annotations is lower for the same recall value. This indicates that the difficulty level of these annotations is higher compared to the original dataset. Fig. 7 shows that training on

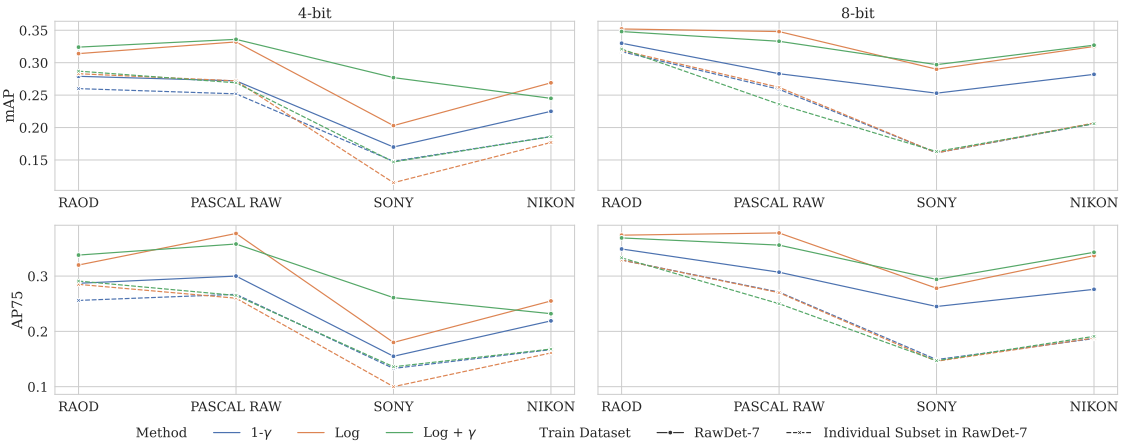

Figure 7: Results for training on combined RAWDET-7 when evaluated on each subset, vs. training only on a subset and evaluated the subset's test set. Combining improves results on all subsets. The model used is Faster RCNN.

the proposed RAWDET-7 dataset improves performance across individual datasets, compared to training the model individually with every subset. This indicates that combining multi-bit depth raw images during training helps the model generalize well on every subset of data, which further proves the usefulness of RAWDET-7.

While object detection annotations provide precise localization, they remain limited to coarse, class-level characterization of scene content. To assess how much fine-grained information is preserved through RAW preprocessing, object descriptions focus on the extent to which object-level semantic, spatial, and contextual details are retained relative to sRGB. These annotated descriptions provide a consistent reference for evaluation, enabling controlled, fine-grained comparisons across preprocessing methods and RAW variants.

6 Conclusion

RAWDET-7 addresses a key bottleneck in RAW-domain vision by providing a large-scale, standardized, and human-validated benchmark for object detection and object description on quantized RAW images. By consolidating and re-annotating four existing datasets, we create a diverse benchmark spanning varied sensors, lighting conditions, environments, and bit-depth regimes, while substantially improving annotation quality, category coverage, and the labeling of small and partially-occluded instances. Beyond detection, the object-description track extends evaluation from class-level recognition to fine-grained semantic, spatial, and contextual understanding, providing a new way to study how much information different RAW processing pipelines preserve.

RAWDET-7 further enables controlled benchmarking under realistic low-bit sensing constraints. Across standard detectors and a large grounding model, our experiments show that suitable RAW-aware input mappings make heavily quantized RAW highly competitive with, and in some settings superior to, standard sRGB pipelines, while preserving substantially richer object-level information than naïve quantization. Supported by human studies validating both the annotation and description pipelines, we believe RAWDET-7 can serve as a strong foundation for future research in low-bit sensing, quantization-aware perception, and sensor-to-task co-design.

Limitations. RAWDET-7 improves coverage of sensor settings by building on diverse prior datasets, but its coverage of the broader RAW sensor landscape remains limited. The dataset is also imbalanced across sensors and bit depths, with most images at 24-bit. Even if the practical impact is unclear, a more balanced sensor representation is desirable. In principle, additional RAW datasets could be merged into RAWDET-7 and annotated using our pipeline, but expanding annotations is costly because high-quality annotation tools (including ours) are paid.

References

Navya Sonal Agarwal, Jan Philipp Schneider, Kanchana Vaishnavi Gandikota, Syed Muhammad Kazim, John Meshreki, Ivo Ihrke, and Michael Moeller. Direct image classification from fourier ptychographic microscopy measurements without reconstruction, 2025.

Shashank Agnihotri, Jonas Jakubassa, Priyam Dey, Sachin Goyal, Bernt Schiele, Venkatesh Babu Radhakrishnan, and Margret Keuper. A granular study of safety pretraining under model abliteration. In *NeurIPS Workshop Lock-LLM Workshop: Prevent Unauthorized Knowledge Use from Large Language Models*, 2025.

A. Bermak and A. Kitchen. A novel adaptive logarithmic digital pixel sensor. *IEEE Photonics Technology Letters*, 18(20):2147–2149, 2006.

Mark Buckler, Suren Jayasuriya, and Adrian Sampson. Reconfiguring the imaging pipeline for computer vision. In *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, 2017.

Mu Cai, Haotian Liu, Siva Karthik Mustikovela, Gregory P. Meyer, Yuning Chai, Dennis Park, and Yong Jae Lee. Making large multimodal models understand arbitrary visual prompts. In *IEEE Conference on Computer Vision and Pattern Recognition*, 2024.

Kai Chen, Jiaqi Wang, and et al. Mmdetection: Open mmlab detection toolbox and benchmark, 2019.

Gheorghe Comanici, Eric Bieber, Mike Schaekermann, Ice Pasupat, Noveen Sachdeva, Inderjit Dhillon, Marcel Blistein, Ori Ram, Dan Zhang, Evan Rosen, et al. Gemini 2.5: Pushing the frontier with advanced reasoning, multimodality, long context, and next generation agentic capabilities. *arXiv preprint arXiv:2507.06261*, 2025.

Ziteng Cui and Tatsuya Harada. Raw-adapter: Adapting pre-trained visual model to camera raw images. In *European Conference on Computer Vision*, pages 37–56. Springer, 2024.

Duc-Tien Dang-Nguyen, Cecilia Pasquini, Valentina Conotter, and Giulia Boato. Raise: A raw images dataset for digital image forensics. In *Proceedings of the 6th ACM multimedia systems conference*, pages 219–224, 2015.

Steven Diamond, Vincent Sitzmann, Frank Julca-Aguilar, Stephen Boyd, Gordon Wetzstein, and Felix Heide. Dirty pixels: Towards end-to-end image processing and perception. *ACM Transactions on Graphics (TOG)*, 40(3):1–15, 2021.

Mishal* Fatima, Shashank* Agnihotri, Marius* Bock, Kanchana Vaishnavi Gandikota, Kristof Van Laerhoven, Michael Moeller, and Margret Keuper. γ-Quant: Towards Learnable Quantization for Low-bit Pattern Recognition. *DAGM German Conference on Pattern Recognition*, 2025.

Xavier Glorot, Antoine Bordes, and Yoshua Bengio. Deep sparse rectifier neural networks. In *Proceedings of the Fourteenth International Conference on Artificial Intelligence and Statistics*, pages 315–323, Fort Lauderdale, FL, USA, 2011. PMLR.

Rahul Gulve, Navid Sarhangnejad, Gairik Dutta, Motasem Sakr, Don Nguyen, Roberto Rangel, Wenzheng Chen, Zhengfan Xia, Mian Wei, Nikita Gusev, Esther Y. H. Lin, Xiaonong Sun, Leo Hanxu, Nikola Katic, Ameer M. S. Abdelhadi, Andreas Moshovos, Kiriakos N. Kutulakos, and Roman Genov. 39 000-subexposures/s dual-adc cmos image sensor with dual-tap coded-exposure pixels for single-shot hdr and 3-d computational imaging. *IEEE Journal of Solid-State Circuits*, 58(11):3150–3163, 2023.

Agrim Gupta, Piotr Dollar, and Ross Girshick. LVIS: A dataset for large vocabulary instance segmentation. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2019.

Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 770–778, 2016.

Andrey Ignatov, Luc Van Gool, and Radu Timofte. Replacing mobile camera isp with a single deep learning model. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition workshops*, pages 536–537, 2020.

Kang Kim and Hee Seok Lee. Probabilistic anchor assignment with iou prediction for object detection. In *Computer Vision–ECCV 2020: 16th European Conference, Glasgow, UK, August 23–28, 2020, Proceedings, Part XXV 16*, pages 355–371. Springer, 2020.

Tzofi Klinghoffer, Siddharth Somasundaram, Kushagra Tiwary, and Ramesh Raskar. Physics vs. learned priors: Rethinking camera and algorithm design for task-specific imaging. In *2022 IEEE International Conference on Computational Photography (ICCP)*, pages 1–12. IEEE, 2022.

Martin Lefebvre and David Bol. Mantis: A mixed-signal near-sensor convolutional imager soc using charge-domain 4b-weighted 5-to-84-tops/w mac operations for feature extraction and region-of-interest detection. *IEEE Journal of Solid-State Circuits*, 60(3):934–948, 2025.

Xuanyu Lei, Zonghan Yang, Xinrui Chen, Peng Li, and Yang Liu. Scaffolding coordinates to promote vision-language coordination in large multi-modal models. In *Proceedings of the 31st International Conference on Computational Linguistics*, pages 2886–2903, Abu Dhabi, UAE, 2025. Association for Computational Linguistics.

Zhihao Li, Ming Lu, Xu Zhang, Xin Feng, M. Salman Asif, and Zhan Ma. Efficient visual computing with camera raw snapshots. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 46(7):4684–4701, 2024a.

Zhihao Li, Ming Lu, Xu Zhang, Xin Feng, M Salman Asif, and Zhan Ma. Efficient visual computing with camera raw snapshots. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 46(7):4684–4701, 2024b.

Tsung-Yi Lin, Michael Maire, Serge Belongie, James Hays, Pietro Perona, Deva Ramanan, Piotr Dollár, and C Lawrence Zitnick. Microsoft coco: Common objects in context. In *Computer Vision–ECCV 2014: 13th European Conference, Zurich, Switzerland, September 6-12, 2014, Proceedings, Part V 13*, pages 740–755. Springer, 2014.

Tsung-Yi Lin, Priya Goyal, Ross Girshick, Kaiming He, and Piotr Dollár. Focal loss for dense object detection. In *Proceedings of the IEEE international conference on computer vision*, pages 2980–2988, 2017.

Anqi Liu, Shiyi Mu, and Shugong Xu. A learnable color correction matrix for raw reconstruction. In *British Machine Vision Conference*, 2024a.

Shilong Liu, Zhaoyang Zeng, Tianhe Ren, Feng Li, Hao Zhang, Jie Yang, Qing Jiang, Chunyuan Li, Jianwei Yang, Hang Su, et al. Grounding dino: Marrying dino with grounded pre-training for open-set object detection. In *European Conference on Computer Vision*, pages 38–55. Springer, 2024b.

Ze Liu, Yutong Lin, Yue Cao, Han Hu, Yixuan Wei, Zheng Zhang, Stephen Lin, and Baining Guo. Swin transformer: Hierarchical vision transformer using shifted windows. In *Proc. of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2021.

William Ljungbergh, Joakim Johnander, Christoffer Petersson, and Michael Felsberg. Raw or cooked? object detection on raw images. In *Scandinavian Conference on Image Analysis*, pages 374–385. Springer, 2023.

Igor Morawski, Yu-An Chen, Yu-Sheng Lin, Shusil Dangi, Kai He, and Winston H Hsu. Genisp: Neural isp for low-light machine cognition. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 630–639, 2022.

Ali Mosleh, Avinash Sharma, Emmanuel Onzon, Fahim Mannan, Nicolas Robidoux, and Felix Heide. Hardware-in-the-loop end-to-end optimization of camera image processing pipelines. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 7529–7538, 2020.

A Omid-Zohoor, D Ta, and B Murmann. Pascalraw: raw image database for object detection, 2014.

Shaoqing Ren, Kaiming He, Ross Girshick, and Jian Sun. Faster r-cnn: Towards real-time object detection with region proposal networks. *IEEE transactions on pattern analysis and machine intelligence*, 39(6):1137–1149, 2016.

Nicolas Robidoux, Luis E Garcia Capel, Dong-eun Seo, Avinash Sharma, Federico Ariza, and Felix Heide. End-to-end high dynamic range camera pipeline optimization. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 6297–6307, 2021.

Olga Russakovsky, Jia Deng, Hao Su, Jonathan Krause, Sanjeev Satheesh, Sean Ma, Zhiheng Huang, Andrej Karpathy, Aditya Khosla, Michael Bernstein, Alexander C. Berg, and Li Fei-Fei. ImageNet Large Scale Visual Recognition Challenge. *International Journal of Computer Vision (IJCV)*, 115(3):211–252, 2015.

Eli Schwartz, Raja Giryes, and Alex M. Bronstein. Deepisp: Toward learning an end-to-end image processing pipeline. *IEEE Transactions on Image Processing*, 28(2):912–923, 2019.

Hendrik Sommerhoff, Shashank Agnihotri, Mohamed Saleh, Michael Moeller, Margret Keuper, and Andreas Kolb. Differentiable sensor layouts for end-to-end learning of task-specific camera parameters. *arXiv preprint arXiv:2304.14736*, 2023.

Ilya Sutskever, James Martens, George Dahl, and Geoffrey Hinton. On the importance of initialization and momentum in deep learning. In *Proceedings of the 30th International Conference on Machine Learning*, pages 1139–1147, Atlanta, Georgia, USA, 2013. PMLR.

David Wan, Jaemin Cho, Elias Stengel-Eskin, and Mohit Bansal. Contrastive region guidance: Improving grounding in vision-language models without training. In *European Conference on Computer Vision*, pages 198–215. Springer, 2024.

Zhenwei Wu, Xinfa Wang, Meng Jia, Minghao Liu, Chengxiu Sun, Chenyang Wu, and Jianping Wang. Dense object detection methods in raw uav imagery based on yolov8. *Scientific reports*, 14(1):18019, 2024.

Ruikang Xu, Chang Chen, Jingyang Peng, Cheng Li, Yibin Huang, Fenglong Song, Youliang Yan, and Zhiwei Xiong. Toward raw object detection: A new benchmark and a new model. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 13384–13393, 2023.

Jianwei Yang, Hao Zhang, Feng Li, Xueyan Zou, Chunyuan Li, and Jianfeng Gao. Set-of-mark prompting unleashes extraordinary visual grounding in gpt-4v, 2023a.

Lingfeng Yang, Yueze Wang, Xiang Li, Xinlong Wang, and Jian Yang. Fine-grained visual prompting. *Advances in Neural Information Processing Systems*, 36:24993–25006, 2023b.

Zhaoyang Yin, Yibing M. Wang, and Eric R. Fossum. Low bit-depth adcs for multi-bit quanta image sensors. *IEEE Journal of Solid-State Circuits*, 56(3):950–960, 2021.

Yin Zhang, Yongqiang Zhang, Zian Zhang, Man Zhang, Rui Tian, and Mingli Ding. Isp-teacher: Image signal process with disentanglement regularization for unsupervised domain adaptive dark object detection. In *Proceedings of the AAAI Conference on Artificial Intelligence*, pages 7387–7395, 2024.

Xiangyu Zhao, Yicheng Chen, Shilin Xu, Xiangtai Li, Xinjiang Wang, Yining Li, and Haian Huang. An open and comprehensive pipeline for unified object grounding and detection. *arXiv preprint arXiv:2401.02361*, 2024.

RAWDet-7: A Multi-Scenario Benchmark for Object Detection and Description on Quantized RAW Images

Supplementary Material

A Experimental Details **1**

A.1 Object Detection Training Details 1

A.2 Details of Foundation Model used for Annotations 1

B User Study Comparing Our New Object Detection Annotations to Old **2**

C Visualizing And Comparing Annotations From RAWDet-7 **2**

D Additional Dataset Statistics **2**

E Additional Benchmarking Details e.g. details on the ISP used for RAW RGB conversion of the images from the RAOD dataset. **5**

F Advantage of Consolidated Dataset **5**

G Results with 4-channel input: RGGB **7**

H Object Descriptions **7**

H.1 Generating Object Descriptions 7

H.2 User Study to Validate Quality of the Object Descriptions 9

H.3 Metrics for Evaluating Object Descriptions 9

H.4 User Study to Verify LLM as a Judge for Metrics 12

A Experimental Details

Following, we provide more details regarding the object detection experimental setup: training details for the architecture and details on using the foundation model for generating the new annotations for RAWDet-7.

A.1 Object Detection Training Details

Following previous works, for training traditional object detection models, we follow multi-scale training where the shorter image side is scaled to one of the sides randomly selected from a set of sides: [480, 512, 544, 576, 608, 640, 672, 704, 736, 768, 800] using nearest neighbor interpolation and the longer side is scaled to maintain the aspect ratio. We apply a warm-up for the first 1000 iterations, linearly increasing the learning rate (lr) from $1e^{-3}$ to $2.5e^{-3}$ followed by a multi-step learning rate scheduler.

A.2 Details of Foundation Model used for Annotations

We used the paid API version at: `https://deepdataspace.com/request_api`. We use their model "DetectionModel.GDino1_5_Pro" for generating the bounding boxes and classes such that we only annotate objects with 20% or less overlap with another object of the same class (following recommended settings).

Table 3: Characteristics of Annotated Objects in RAWDet-7.

	Bicycle	Bus	Car	Motorcycle	Person	Tram	Truck
Avg. Obj Size (H × W)	477 × 431	224 × 301	250 × 257	231 × 195	508 × 213	177 × 307	231 × 257
Avg. Obj Count/Image	1.95	1.28	7.63	2.12	3.62	1.12	1.53
Avg. Aspect Ratio	0.939	1.486	1.350	0.888	0.528	2.192	1.197

The text prompt we used for generating the annotations was "car . truck . tram . person . bicycle . motorcycle . bus .". Each query to the API costs ¥0.1, thus, including the debugging and ablations over prompts and classes, the total cost was ¥3402.80. Finally, after obtaining the annotations, we observed that the model was often hallucinating low-confidence objects. Thus, we filter out these hallucinated objects, after carefully observing multiple annotations and confidences, by using a confidence score threshold of 0.8, i.e. only annotations with confidence scores of 0.8 and above were retained, and the remaining were filtered out.

B User Study Comparing Our New Object Detection Annotations to Old

We did a user survey for our new data annotations and received 53 responses. For the survey, we collected 100 random pairs of annotations such that for each image chosen at random, we had the annotation from RAWDet-7 and the annotation provided by the original dataset. We split the 100 random pairs of annotations into 4 sets of 25 questions each. Each question shuffled the order of option A and option B, with annotation from RAWDet-7 for that image being either option A or option B, and the other option being the annotation from the original dataset.

Given 25 questions to each user, and since 53 users responded, we had exactly 1325 instances of comparisons between the RAWDet-7 annotations and the original annotations answered. From these 1325 instances, users preferred the RAWDet-7 annotations in 1002 instances. That is, **users preferred** RAWDet-7 **annotations for 75.62% of all the instances**. This clearly demonstrates the improvement of the annotations provided by RAWDet-7. For ease of access to the comparisons, in Section C, we extended the comparison from Figure 1 and show a few examples comparing the annotations from RAWDet-7 and the original annotations.

Limitations Of The User Study. Post the study, a few users informed that they had to zoom in quite a bit to confirm that one annotation is better than the other. We suspect that the annotations that required zooming in were the annotations from RAWDet-7 since it provides very fine-grained detections as well. Users reported that, in a few instances, those who are not vigilant may not choose the RAWDet-7 annotations, and thus the final scores might underrepresent the benefit of the annotations from RAWDet-7.

C Visualizing And Comparing Annotations From RAWDet-7

In Figure 9, we observe that RAWDet-7 covers many small objects in the images that were originally missing from the annotations, and correctly labels many misclassified objects in the original annotations.

D Additional Dataset Statistics

In Table 3, we list different object characteristics in the proposed RAWDet-7dataset. Average object size is computed by averaging the height and width of bounding boxes across all images in the train and validation sets for a particular category. We observe that the bicycle category has the largest object size, and the motorcycle, truck, and tram have the smallest. The category "car" has the highest average object count per image, which is intuitive due to the outdoor nature of the proposed dataset. Average aspect ratio is computed by calculating the aspect ratio (width/height) of every object in a category and averaging over the entire dataset.

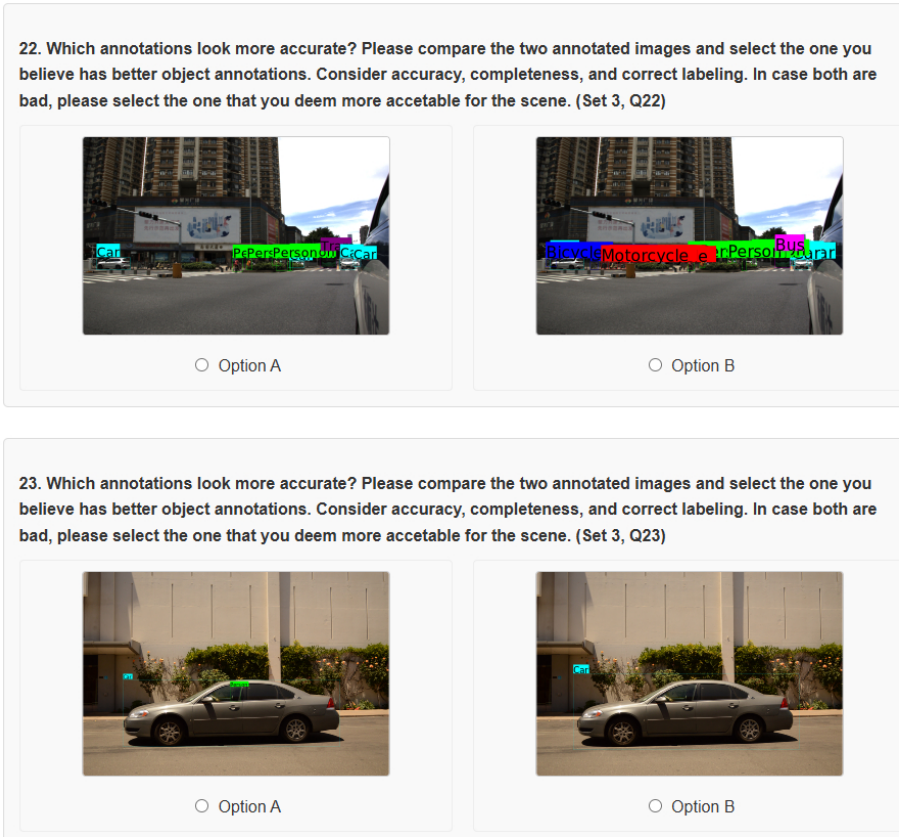

Figure 8: Examples from the annotation-quality user study, where participants inspected bounding boxes and compared newly generated annotations against older versions to assess improvements in quality.

Figure 9: Comparing ground truth annotations provided in the original datasets and the new ones proposed in RAWDet-7 as used in the user study. Please note that the visualizations here are heavily compressed due to size limitations.

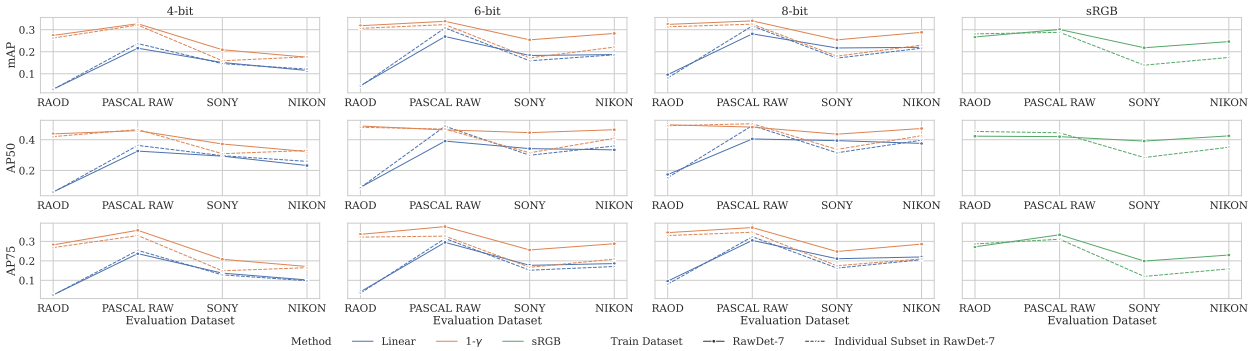

Figure 10: Results with PAA for training on combined RAWDet-7 when evaluated on each subset, vs. training only on a subset and evaluated on the subset's test set. Combining improves results on the majority of the subsets in the case of gamma scaling and sRGB. These comparisons include all metrics (mAP, AP50, AP75) and cover 4, 6, and 8-bit depth data as well as sRGB.

E Additional Benchmarking Details e.g. details on the ISP used for RAW RGB conversion of the images from the RAOD dataset.

Except for the Zurich Dataset, the remaining individual datasets have a Bayer pattern of RGGB, whereas Zurich raw images have an RGBG pattern. We extract red, green, and blue channels from the relevant Bayer pattern and take the mean of the double green channels. For quantization experiments, we use *torch.floor* as a quantization function. In the case of gamma scaling, we use a straight-through estimator to let the gradients pass through the quantization operation. The RGB images of all the individual datasets are publicly provided along with the raw images, except for the RAOD dataset. So, we generated them by first extracting the Red, Green, and Blue channels from the original raw image, followed by a gray world white balance algorithm. Then, we apply gamma correction, where the value of gamma is chosen by hit-and-run to be 0.09.

F Advantage of Consolidated Dataset

We show the advantage of adding all the individual datasets together for training vs. training only with the individual datasets in Figure 10 for PAA architecture. We observe that while training with linear quantization without any scaling results in no performance gain when trained in a combined fashion, gamma scaling and RGB experiments usually benefit from combined training. This is especially visible in the case of the Sony and Nikon Datasets.

In Figure 11, we show results with FRCNN for training on combined RAWDet-7 when evaluated on each subset, vs. training only on a subset and evaluated on the subset's test set. Combining improves results on the majority of the subsets in the case of gamma scaling and sRGB. These comparisons include all metrics (mAP, AP50, AP75) and cover 4, 6, and 8-bit depth data as well as sRGB.

In Figure 12, we show precision-recall curves for the RAOD dataset. "Old" refers to the annotations as proposed in the original dataset, whereas "New" refers to the annotations in RAWDet-7. All the models are Faster R-CNNs, trained for 8-bit quantization and jointly learnt with 1 gamma. As mentioned in the legend, the keys are "(Train Dataset - Evaluation Dataset)". Curves are shown for the classes car, truck, tram, and person.

In **??**, we summarize the statistics on the different datasets that we consolidate into RAWDet-7. The individual datasets have each a particular sensor model and bit depth. The combined dataset, therefore, provides significantly increased data diversity.

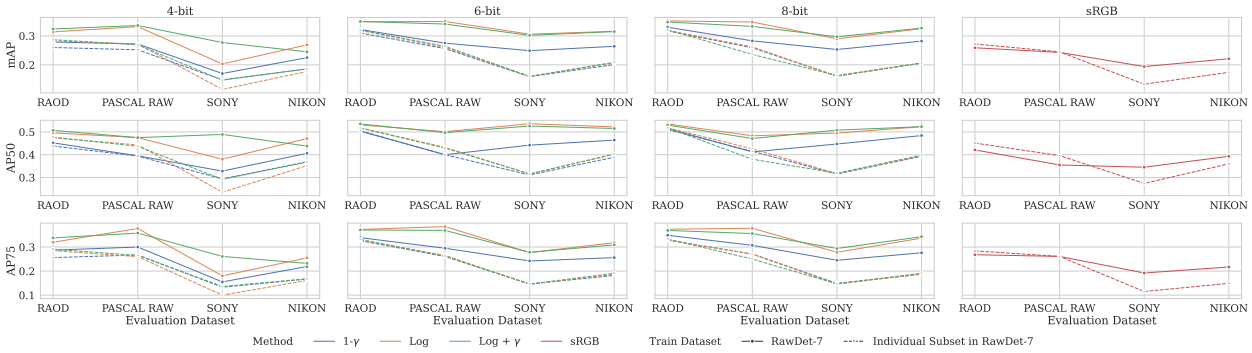

Figure 11: Results with FRCNN for training on combined RAWDet-7 when evaluated on each subset, vs. training only on a subset and evaluated on the subset's test set. Combining improves results on the majority of the subsets in the case of gamma scaling and sRGB. These comparisons include all metrics (mAP, AP50, AP75) and cover 4, 6, and 8-bit depth data as well as sRGB.

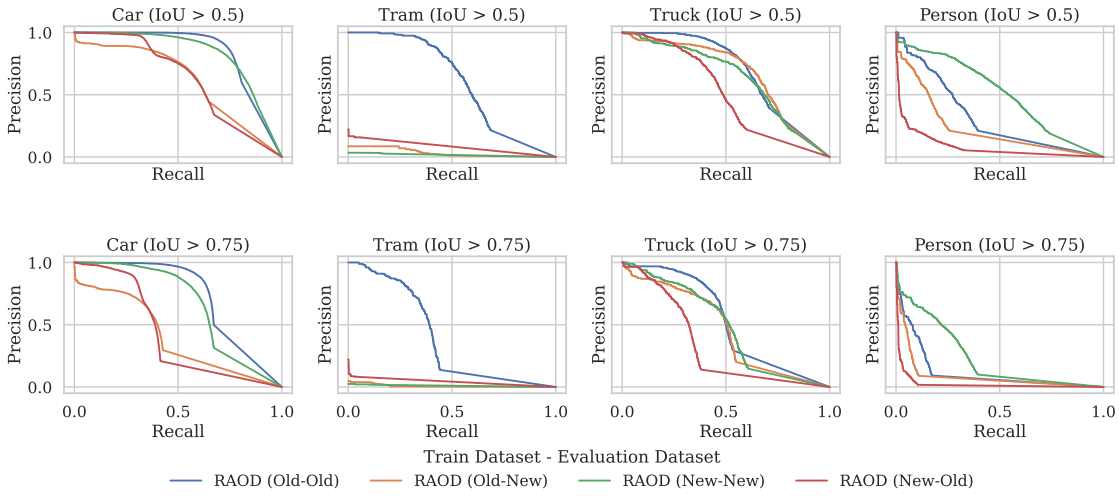

Figure 12: Precision Recall Curves for RAOD. "Old" refers to the annotations as proposed in the original dataset, whereas "New" means the annotations in RAWDet-7. All the models are trained on Faster R-CNN for 8-bit quantization and jointly learnt with 1 gamma. As mentioned in the legend, the keys are "(Train Dataset - Evaluation Dataset)". Curves are shown for the classes car, truck, tram, and person.

In Figure 13, we provide detailed dataset statistics. In particular, we report statistics on instances in the original proposed datasets indicated by 'OLD' and the re-annotated images for those respective datasets indicated by 'NEW' for the classes in RAWDet-7. The proposed RAWDet-7 is a consolidation of RAW input images from the 'OLD' datasets provided with the 'NEW' annotations. For some classes, it appears that 'OLD RAOD' has more instances than the proposed RAWDet-7. Note that RAOD has a large number of objects annotated. However, as shown in Figure 1, the original annotations of the RAOD dataset contain hallucinations for classes like 'Tram', 'Truck', and 'Car'.

The consolidated dataset RAWDet-7 provides cleaned labels for the diverse RAW data samples.

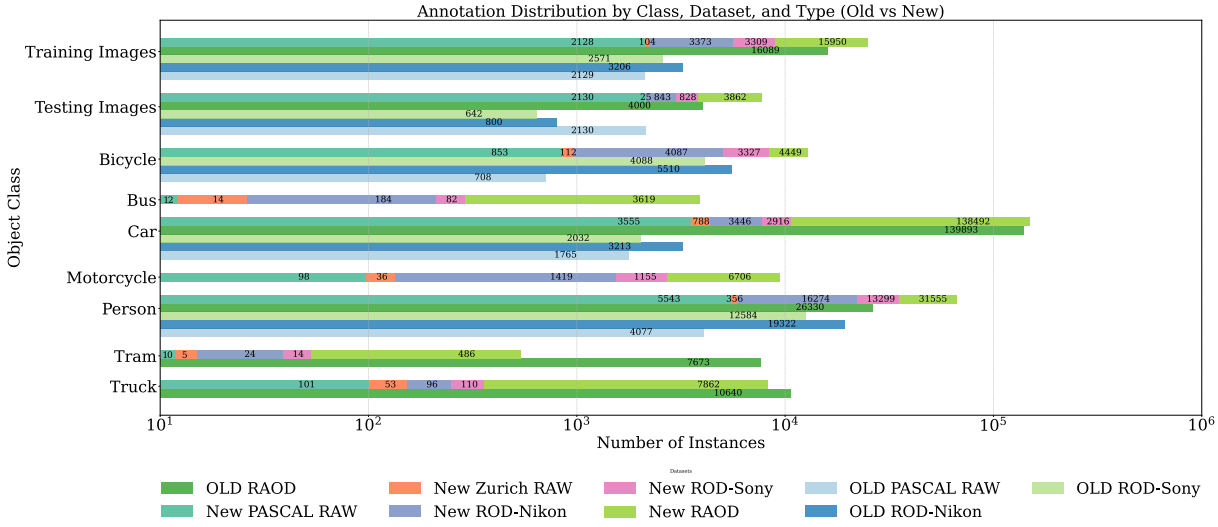

Figure 13: Dataset Statistics: Breakdown of instances in the original proposed datasets indicated by 'OLD' and the re-annotated images for those respective datasets indicated by 'NEW' for the classes in RAWDet-7. The proposed RAWDet-7 is a consolidation of RAW input images from the 'OLD' datasets provided with the 'NEW' annotations. For some classes, it appears that 'OLD RAOD' has more instances than the proposed RAWDet-7. However, as shown in Fig. 1, the original annotations of the RAOD dataset contain hallucinations for classes like 'Tram', 'Truck', and 'Car'.

Table 4: Detection performance (mAP, mAP_{50}, and mAP_{75}) under different bit-depth and color encoding settings.

Setting	**mAP** ↑	**mAP$_{50}$** ↑	**mAP$_{75}$** ↑
4-bit RGB	30.3	48.5	31.6
4-bit RGGB	29.0	48.3	30.1
6-bit RGB	35.1	54.5	37.1
6-bit RGGB	34.0	54.5	35.7
8-bit RGB	36.1	55.8	38.3
8-bit RGGB	34.3	55.0	35.9

G Results with 4-channel input: RGGB

We conduct experiments (see Table 4) by extracting the four Bayer channels (R, G, G, B) from the raw images and modifying Faster R-CNN to accept 4-channel inputs. Both RGGB and RGB experiments were trained jointly with a 1-gamma setting for 140 epochs. We observe that the RGB and RGGB inputs perform on par for various bit depth settings. Please note that this on-par performance is despite the fact that we used a ResNet-50 backbone pretrained on ImageNet, which is optimized for 3-channel RGB images. However, we find this result also intuitive since the information provided to the network in RGB and RGGB is very similar, and only a little extra resolution is provided by RGGB.

H Object Descriptions

H.1 Generating Object Descriptions

For object captioning, we use the `caption prompt used for object-level descriptions` prompt with `Gemini-2.5-Pro`. The model receives the overlaid image where each detected object is marked with a

numbered square, together with the list of corresponding class labels. The prompt enforces a strict line-based output format of the form "`<number>: <caption>`" and requires exactly one caption per numbered object, in order from 1 to N. We call `Gemini-2.5-Pro` on each marked image (twice for the sRGB reference and once for each RAW or RGB-downsampled variant) and then parse the returned text using a regular expression to obtain a dictionary that maps object indices to their captions, which is then used for both analysis and visualization.

H.1.1 Prompts for Generating Object Descriptions

Caption prompt used for object-level descriptions

```
You will be shown a photo with multiple objects with these 7 classes of interest: cars, truck,
    tram, person, bicycle, motorcycle, bus.
Each object of interest is marked by a black square containing a coloured number.

Task: For EACH object with a marked number, produce ONE detailed, factual caption that ONLY
    describes that numbered object.
Include concrete details when visible: color, material/texture, approximate size class, local
    position (e.g., top-left),
distinctive parts, visible text/markings, and immediate context relations.

STRICT OUTPUT:
Write ONE line per object using this exact pattern: <number>: <caption>
Acceptable separators after the number are ':', ')', '.', or '-' (e.g., '1: ...' or '1) ...').
Do NOT include headings, bullets, blank lines, or any extra text before/after the list.
If a numbered object cannot be seen or identified, write exactly: 'I cannot see the object'.
The numbering starts at 1 and goes up to N, where N is the total number of marked objects in
    the image.
There should be exactly N lines in your output, one for each numbered object.
If N is the highest number visible, ensure you include captions for all numbers from 1 to N in
    order.
We will tell you the object class of each marked object in the following way:
1: <CLASS>
2: <CLASS>
...
N: <CLASS>

If a numbered mark itself cannot be seen or identified in the image, write exactly: 'I cannot
    see the mark'.
DO NOT HALLUCINATE ANY MARKED NUMBERS.
DO NOT TALK ABOUT THE MARK SQUARES OR NUMBERS ITSELF or the WHOLE IMAGE.
```

H.2 User Study to Validate Quality of the Object Descriptions

We conducted a user study to evaluate the effectiveness of generated captions. We selected 100 images with the highest mean detail levels and then randomly sampled 20 images and their respective captions from these for each user study. We chose the 100 images with the highest mean detail level, since usually it is the longer output text where generative models hallucinate the most, as they often try to make up details. We wanted the user study to validate that this is not the case in our proposed object descriptions. Participants were shown each image along with its captions and asked to rate the quality of the captions on how well they described the marked objects in each image on a scale from 0 to 5. Here, 0 meant that the object description does not describe the object at all, and 5 meant that the object description describes the respective marked object reasonably well, leaving out almost no describable visual cues.

A total of 63 users participated in this study, so a total of 63×20, that is, 1260 samples were collected for the quality of the captions. We received an average rating of $\frac{4.07\pm0.48}{5}$, that is $\approx$82% score, which demonstrates the good quality of the object descriptions generated.

H.3 Metrics for Evaluating Object Descriptions

- **BLEU**: Measures how much the generated text overlaps in wording with a reference.
- **Regex Match**: Checks if the output matches a specific pattern or format.

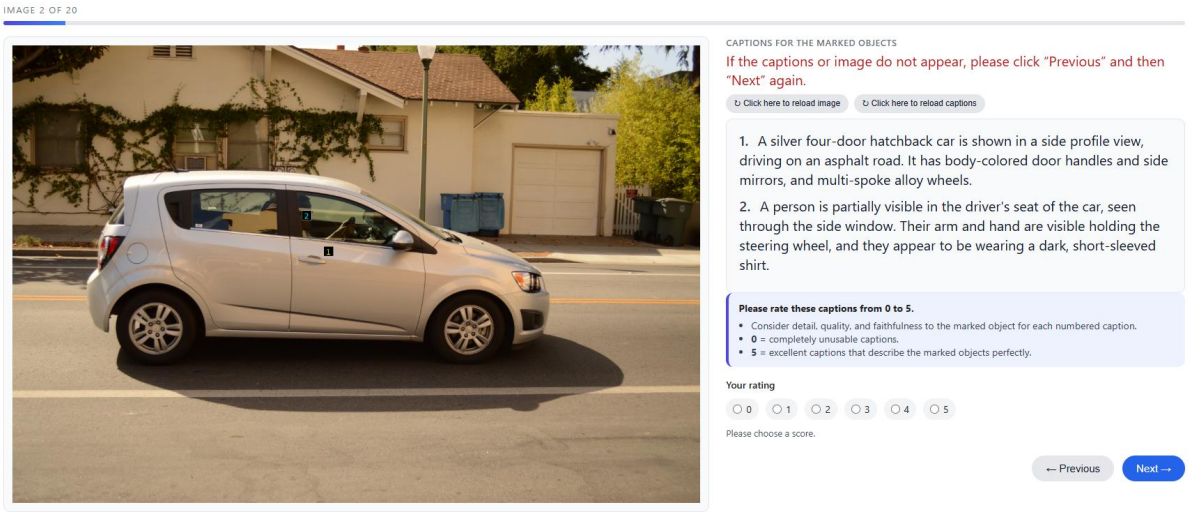

Figure 14: Example from the user study used to assess caption quality. Participants were shown images with designated markers and asked to rate the quality of the accompanying captions, where 0 is completely unusable and 5 is fully descriptive, clear, and highly informative.

- **Similarity (1–10)**: Rates how semantically similar the meaning of two texts are, irrespective of the specific details.
- **Detail Level (0–10)**: Rates how much detail or specificity the output contains.
- **Detail Match (1–10)**: Rates how well specific details in the output match the reference.

Similarity. To quantify semantic similarity between two captions that describe the same object, we use the `Similarity scoring prompt` with `Gemini-2.5-Flash-Lite`. This prompt presents Caption A and Caption B and instructs the model to ignore surface-level differences such as wording (for example, "car" versus "sedan") and colour or brightness variations, and to focus instead on whether both captions refer to the same underlying object with consistent meaning. The model is required to output a single integer in [1, 10] on a separate line, where 1 denotes unrelated or contradictory descriptions and 10 denotes near-identical semantics. We use this score as the *Similarity* metric for each caption pair, where one of the captions is always the first caption generated for the full-scale RGB image.

Detail Level. To assess how informative a single caption is, we use the `Detail level prompt` with `Gemini-2.5-Flash-Lite`. The prompt asks the model to rate the richness of grounded detail in one caption only, again returning a single integer score in [0, 10]. The instructions explicitly direct the model to base its judgment on concrete attributes such as materials, positions, parts, readable text, and local relations, while ignoring fluency and style. A score of 0 indicates no grounded detail, whereas 10 corresponds to a very specific, highly descriptive caption. We treat this as the *Detail Level* metric and compute it for each caption that we evaluate.

Detail Match. Finally, we measure the consistency of fine-grained information between two captions using the `Detail match prompt`, again with `Gemini-2.5-Flash-Lite`. Here, Caption A (first full-scale sRGB caption) acts as the reference and Caption B as the candidate, and the model is instructed to assign a score in [1, 10] based on the overlap of concrete, verifiable details such as materials, positions, parts, readable text and relations, while explicitly disregarding differences in colours or brightness due to varying image processing. Low scores correspond to almost no shared details, whereas scores near 10 indicate high

overlap in grounded content. This yields the *Detail Match* metric, which we aggregate over objects and variants to analyze how well different pipelines preserve detailed object-level information.

H.3.1 Prompts for LLM as a Judge

Similarity scoring prompt

```
You are scoring the semantic similarity between two captions that describe the SAME object.
Check if they are roughly talking about the same object, ignore details.
Disregard similarity in colour, and disregard difference in brightness of the colours;
    disregard the exact word used, for example, car or sedan; focus on overall meaning match.
Return ONLY one integer in [1,10] on a single line with no other text.
Scoring: 1=unrelated/contradictory, 10=near-identical meaning (paraphrases penalize
    contradictions).
Be strict about the overall meaning match.

Caption A:
{A}

Caption B:
{B}
```

Detail level prompt

```
You are scoring the richness of a SINGLE caption, that is, how long and detailed and
    descriptive it is.
Return ONLY one integer in [0,10] on a single line with no other text.
Score based on concrete grounded attributes (colors, materials, positions, parts, readable text
    , relations).
Be strict about verifiable details only. 10=very rich and specific, 0=no grounded detail.
Do not look for the length of the caption, only the amount of concrete details.
Ignore fluency/style. 0=no grounded detail, 10=very rich and specific.

Caption:
{C}
```

Detail match prompt

```
You are scoring the overlap of concrete details between two captions.
Return ONLY one integer in [1,10] on a single line with no other text.
Score high only if both share consistent verifiable details (materials, positions, parts,
    readable text, relations).
Disregard colour differences, and difference in brightness of the colours as the images are
    processed differently, and colour differences are expected, so disregard them.
Ignore fluency/style. 1=no shared details, 10=high detail overlap.
Be strict about concrete detail match.

Caption A:
{A}

Caption B:
{B}
```

H.4 User Study to Verify LLM as a Judge for Metrics

In order to validate the usage of LLMs as a judge to measure description similarity and level of detail, we also conducted a user study. Figure 15 provides an example from this user study. We gather a random subset of 50 images with the two object descriptions per marked object in the images, gathered from the high-resolution sRGB version of each image, and ask users to provide scores between 0 and 10 for the description similarity, the amount of details provided, and the agreement of the given details.

Each user rates 10 of the 50 randomly sampled images at a time. We collected 9 such user studies, so 90 images and their respective pair of object descriptions for each marked object were rated, that is, a total of 391 caption pairs. We calculate the correlation in the mean similarity, mean detail level, and mean detail match between Gemini-2.5-Flash-Lite as a judge and a human as a judge from each user study. For similarity, we get a correlation of 0.61, for the detail level, we get a correlation of 0.71, and for the detail match, we get a correlation of 0.58. There are all high positive correlations, demonstrating that Gemini-2.5-Flash-Lite can be used as a judge for comparing two object descriptions at a time. Please note that using a better LLM like ChatGPT5 or Gemini-2.5-Pro as a judge might lead to better correlations with humans as a judge; however, this would incur high costs, and thus using Gemini-2.5-Flash-Lite is the most cost-effective way. Using open-source models for LLM as a judge might seem like a reasonable choice; however, as shown by Agnihotri et al. (2025), open-source models like Qwen3, GPT-oss have significantly low correlation with human judgment compared to closed and paid models.

The specific instructions given to the user, for "human as a judge" are:

Thank you for taking part in this study. In each trial you will see a high-resolution image with numbered marks on several objects. For every numbered object there are two automatically generated captions that are intended to describe the same marked object.
Your task is to rate the relationship between the two captions for each numbered object. For every object number, you will provide one visibility judgment and four scores on a scale from 0 to 10:

- Is the marked object actually visible? (Dependent on the image) Decide whether the object with this number is fully visible in the image, only partially visible, or not visible at all.
- Semantic similarity (Independent of the image) - Do both captions clearly talk about the same object, ignoring fine-grained detail.
- Level of detail of Caption 1 (Independent of the image) - How rich and specific Caption 1 is, in terms of concrete, verifiable details (independent of the image, verifying if the caption itself has obvious conflicts in the text).
- Level of detail of Caption 2 (Independent of the image) - Same as above, now for Caption 2.
- Detail match (Independent of the image) - How well the concrete details in Caption 1 and Caption 2 agree with each other.

Rating scale (0 to 10)

- 0 - No similarity or no grounded details at all.
- 5 - Moderate similarity or a moderate amount of grounded details.
- 10 - Very high similarity or very rich and specific grounded details.

The 0 to 10 scale applies to the four numeric scores. The visibility question instead uses the options Yes, No, and Only partially.

Only for the visibility question (Is the marked object actually visible?), you should look at the image. For all other scores, please ignore the image itself and judge only the two captions for that object. If a caption contains many details that are internally inconsistent or impossible (for example, mentioning both headlights and taillights when only the front of a car can be visible), please give low scores for the relevant detail and detail-match questions.

Please pay attention that each caption is tied to a numbered object in the image. When you score, always think about the object with that number, not the whole image.

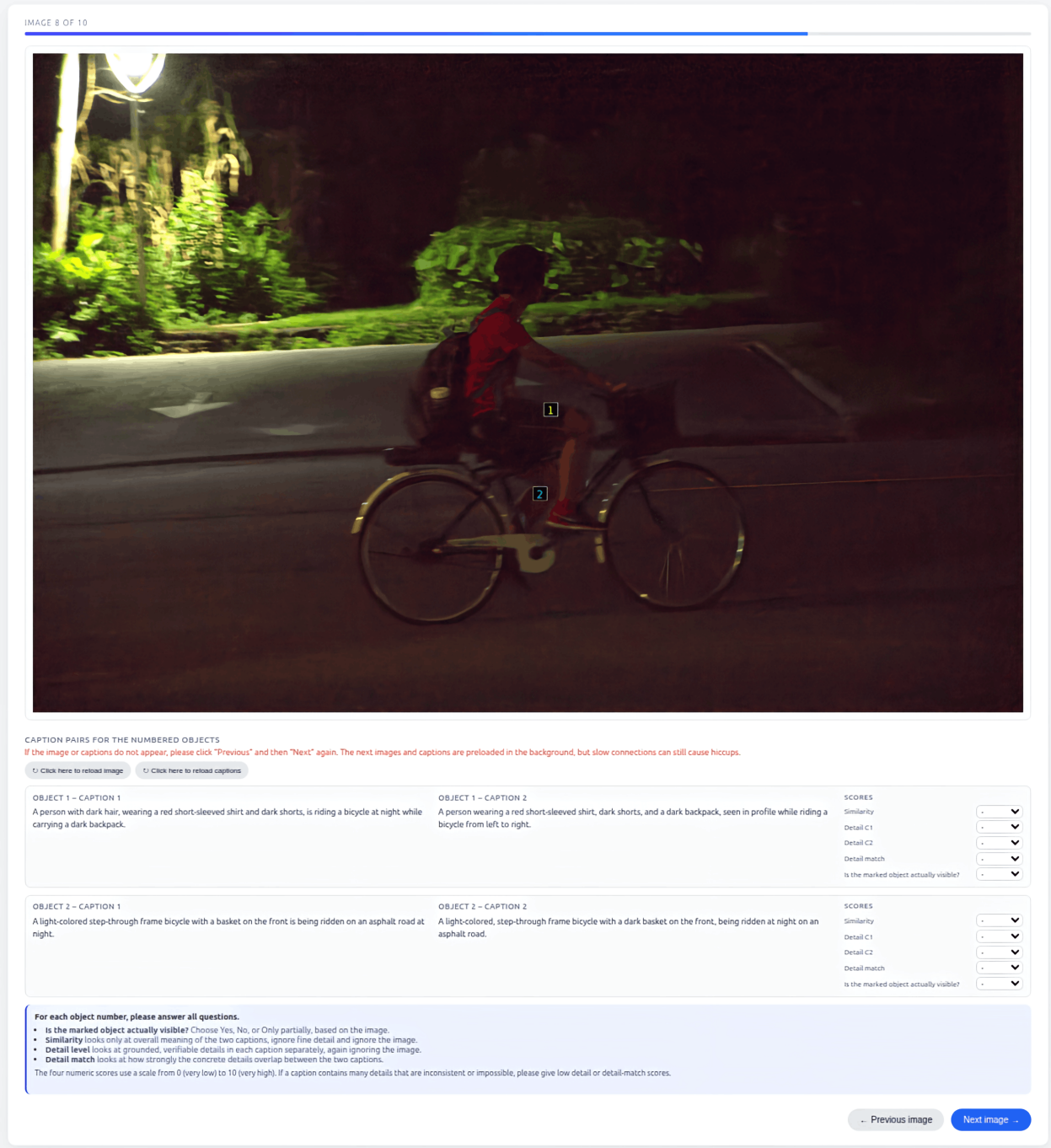

Figure 15: User-study example used to evaluate the performance of an LLM acting as a judge. Participants rated captions based on similarity with one another and richness of details. To assess the captioning model's hallucinations, they also indicated whether the objects described were visible in the images or not.